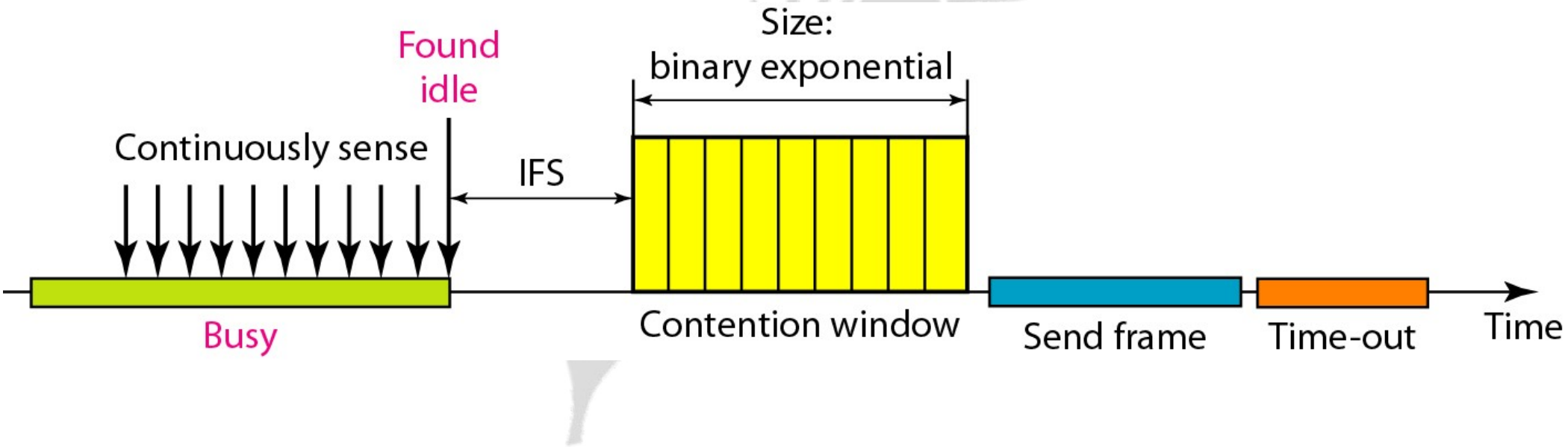
Multiple Access →Random Access →CSMA/CA

- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
- In contrast to CSMA/CD that deals with collisions after their occurrence, CSMA/CA prevents collisions prior to their occurrence
- The basic idea behind CSMA/CA is that the station should be able to receive while transmitting, to detect a collision from different stations
- In wired networks, if a collision has occurred then the energy of received signal almost doubles and the station can sense the possibility of collision
- In case of wireless networks, most of the energy is used for transmission and the energy of received signal increases by only 5 – 10 % if a collision occurs. It can't be used by the station to sense collision. Therefore CSMA/CA has been specially designed for wireless networks

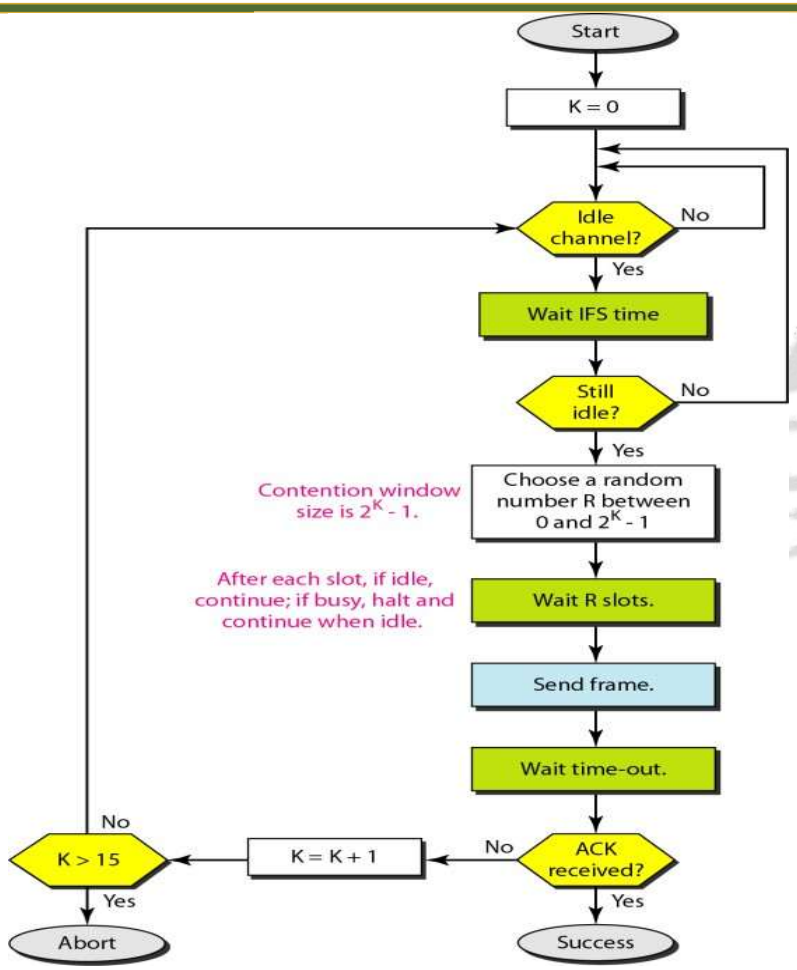
Multiple Access →Random Access →CSMA/CA

- These are three types of strategies:
  - Inter Frame Space (IFS):
    - The concept of an IFS allows wireless systems to implement a simple priority system. Systems with high priority data have shorter IFS times, so whenever the medium is clear they will be the first to send
  - Contention Window:
    - It is the amount of time divided into slots. A station which is ready to send frames chooses random number of slots as wait time
  - Acknowledgements:
    - The positive acknowledgements and time-out timer can guarantee a successful transmission of the frame

CSMA/CA


Found
idle
Size:
binary exponential
Continuously sense
IFS
Busy
Contention window
Send frame
Time-out
Time

CSMA/CA



Channel idle? Don't transmit yet!
Wait IFS time

Still idle after IFS? Don't transmit yet!
Now in Contention Window.
Choose random number and wait that many slots

CSMA/CA

- Algorithm for CSMA/CA
  - When a frame is ready, the transmitting station checks whether the channel is idle or busy
  - If the channel is busy, the station waits until the channel becomes idle
  - If the channel is idle, the station waits for an Inter-frame gap (IFG) amount of time and then sends the frame
  - After sending the frame, it sets a timer
  - The station then waits for acknowledgement from the receiver. If it receives the acknowledgement before expiry of timer, it marks a successful transmission
  - Otherwise, it waits for a back-off time period and restarts the algorithm

CSMA/CD - CA

- Eg:
  - A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal,) is 25.6 μs, what is the minimum size of the frame?

Solution

size of frame = 2 * propagation delay * bandwidth

$= 2 * (25.6 * 10^{-6}) * 10 * 10^{6}$

= 512 bits or 64 bytes

CSMA/CD - CA

- Eg:
  - In a CSMA/CD, network running at 1 Gbps over 1 km cable with no repeaters, the signal speed in the cable is 200000 km/sec. What is minimum frame size?

Solution

$$\text{Propagation delay } (T_p) = \text{Distance/Propagation speed}$$
$$= 1 \text{ km} / (200000 \text{ km/sec})$$
$$= 0.5 * 10^{-5} \text{ sec}$$
$$= 5 * 10^{-6} \text{ sec}$$
$$\text{Frame size} = 2 * \text{Propagation delay} * \text{Bandwidth}$$
$$= 2 * 5 * 10^{-6} * 10^{9}$$
$$= 10000 \text{ bits`}$$

CSMA/CD - CA

- Eg:
  - A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA / CD. The signal travels along the wire at 2 x 10^8 m/sec. What is the minimum packet size that can be used on this network?

Solution

$$\text{Propagation delay } (T_p) = \text{Distance/Propagation speed}$$
$$= 2 \text{ km} / (2 * 10^8 \text{ m/sec})$$
$$= 10^{-5} \text{ sec}$$
$$\text{Frame size} = 2 * \text{Propagation delay} * \text{Bandwidth}$$
$$= 2 * 10^{-5} * 10^7$$
$$= 200 \text{ bits}$$

CSMA/CD - CA

- Eg:
  - A and B are the only two stations on Ethernet. Each has a steady queue of frames to send. Both A and B attempts to transmit a frame, collide and A wins first back off race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second back off race is ____?

CSMA/CD - CA

Solution

In 1st Transmission Attempt:

Both the stations A and B attempts to transmit a frame

collision occurs

Back Off Algorithm runs

Station A wins and successfully transmits its 1st data packet

In 2nd Transmission Attempt:

Station A attempts to transmit its 2nd data packet

Station B attempts to retransmit its 1st data packet

collision occurs

CSMA/CD - CA

Solution

Now, We have to find the probability of station A to transmit its 2nd data packet successfully after 2nd collision

After the 2nd collision,
At Station A:
2nd data packet of station A undergoes collision for the 1st time
So, collision number for the 2nd data packet of station A = 1
Now, station A randomly chooses a number from the range $[0, 2^1-1] = [0, 1]$
Then, station A waits for back off time and then attempts to retransmit its data packet

At Station B:
1st data packet of station B undergoes collision for the 2nd time
So, collision number for the 1st data packet of station B = 2
Now, station B randomly chooses a number from the range $[0, 2^2-1] = [0, 3]$
Then, station B waits for back off time and then attempts to retransmit its data packet

CSMA/CD - CA

Solution

Following 8 cases are possible

From here,

Probability of A winning the 2nd back off race

= 5/8

= 0.625

Station A	Station B	Remark
0	0	Collision
0	1	A wins
0	2	A wins
0	3	A wins
1	0	B wins
1	1	Collision
1	2	A wins
1	3	A wins

12-2 CONTROLLED ACCESS

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.

Topics discussed in this section:

Reservation
Polling
Token Passing

Figure 12.18 *Reservation access method*

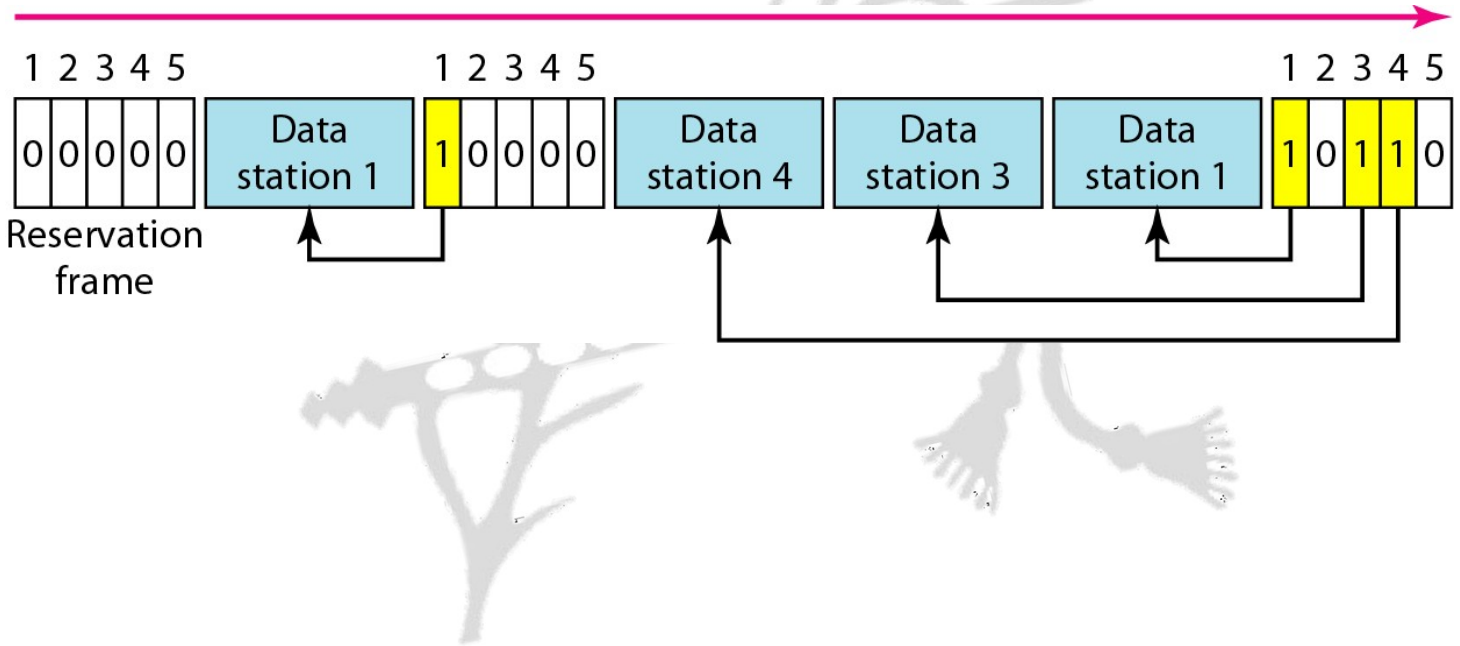


Figure 12.19 *Select and poll functions in polling access method*

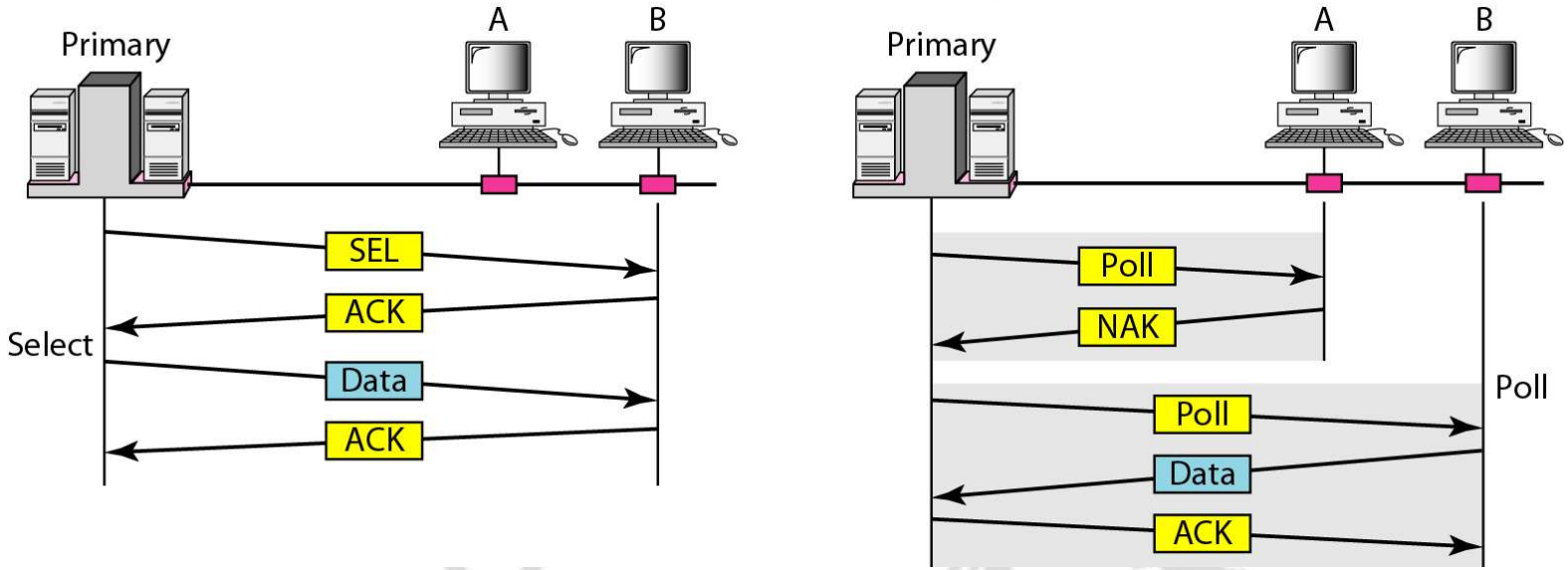
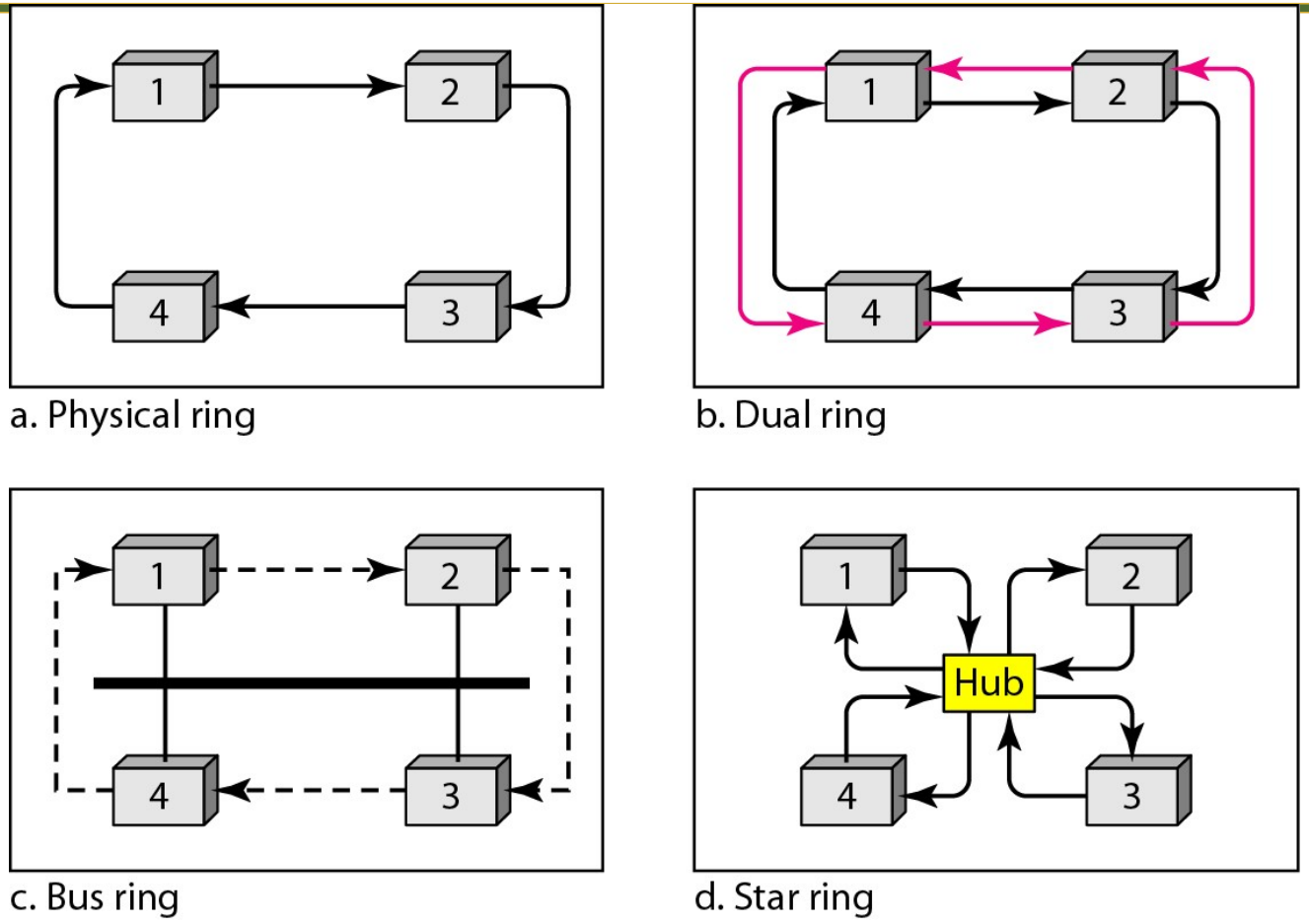


Figure 12.20 *Logical ring and physical topology in token-passing access method*



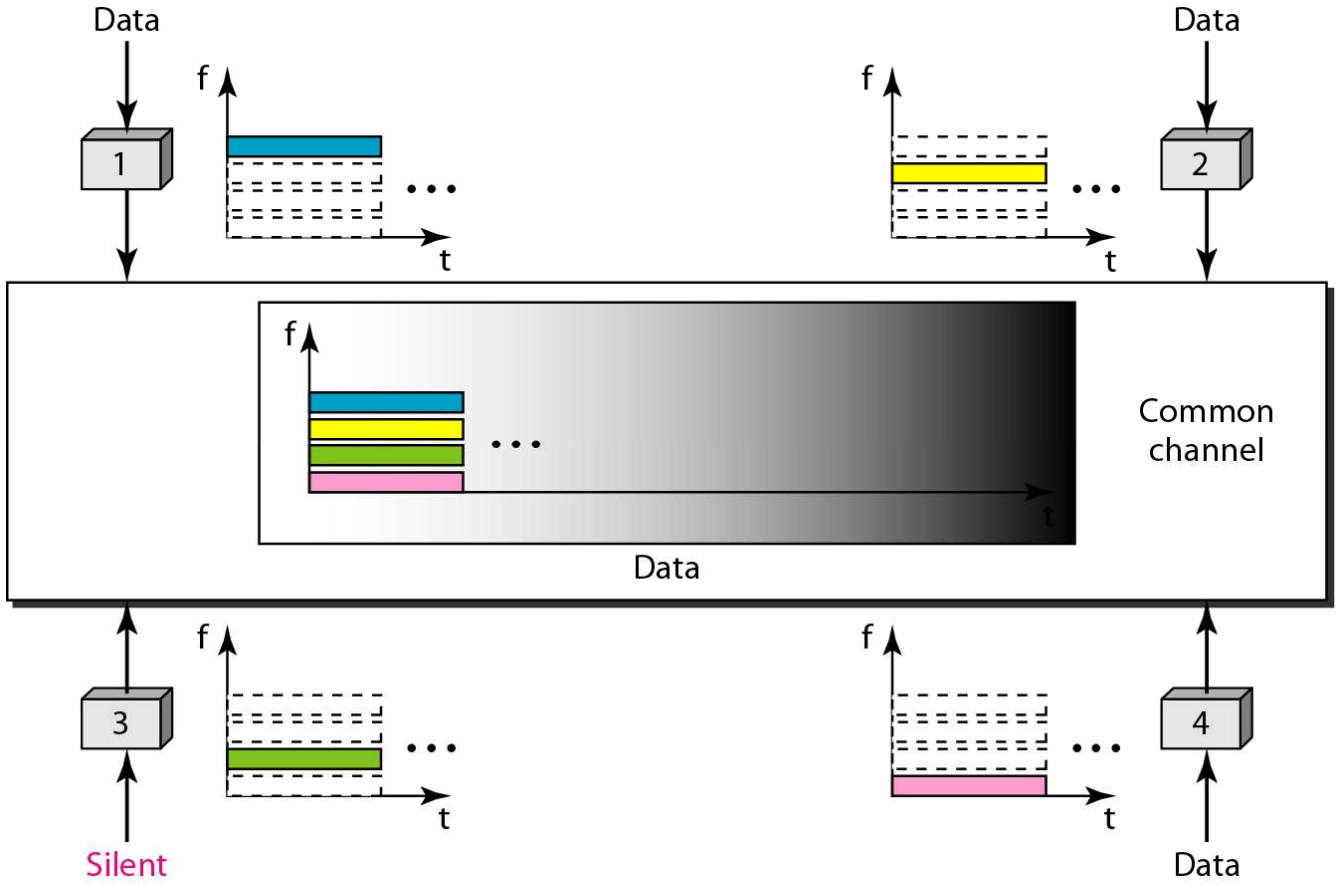
12-3 CHANNELIZATION

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols.

Topics discussed in this section:

Frequency-Division Multiple Access (FDMA)
Time-Division Multiple Access (TDMA)
Code-Division Multiple Access (CDMA)

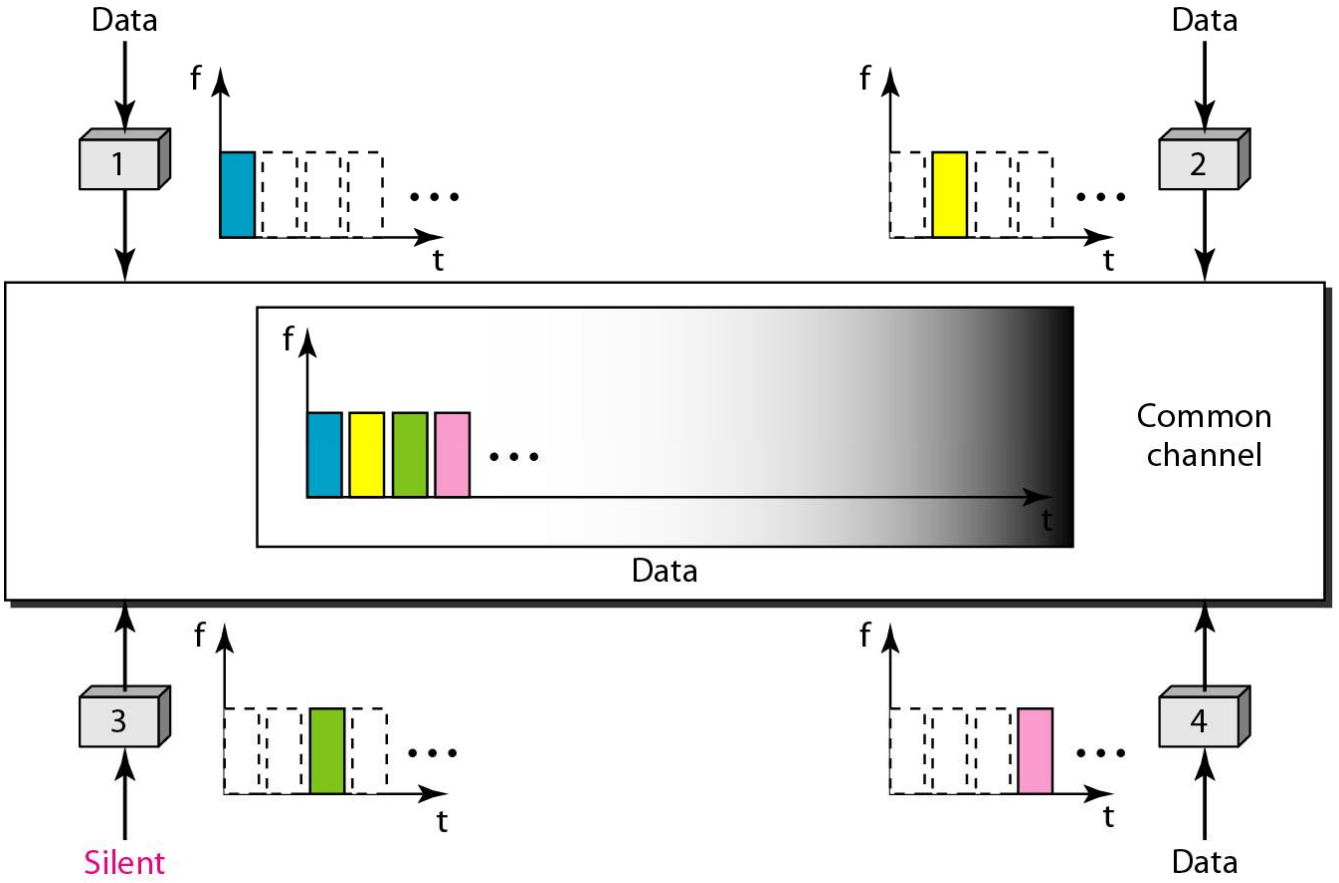
Figure 12.21 *Frequency-division multiple access (FDMA)*



Note

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

Figure 12.22 *Time-division multiple access (TDMA)*



Note

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

Note

In CDMA, one channel carries all transmissions simultaneously.

Figure 12.23 *Simple idea of communication with code*

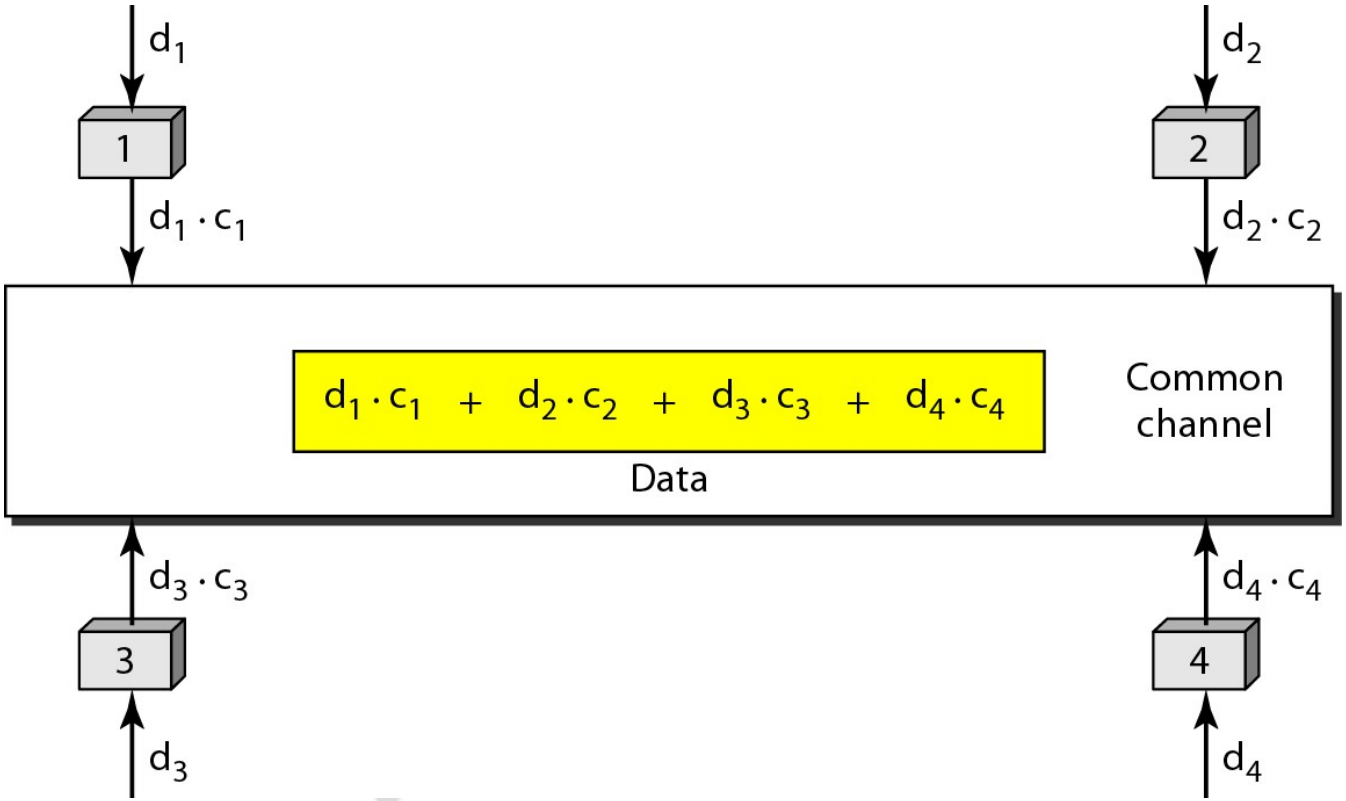


Figure 12.24 *Chip sequences*

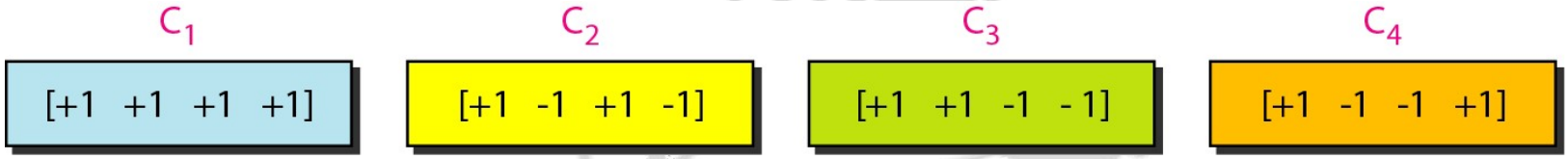


Figure 12.25 *Data representation in CDMA*



Figure 12.26 *Sharing channel in CDMA*

